

2-Infant Morality

Helper / Hinderer Study

MATH 213

Introduction

Previously, we used coin-flipping to simulate what might have happened under the assumption of the null hypothesis, that the children were selecting helpers/hinderers at random.

Here, we run a larger simulation under the assumption of the same null hypothesis, using the computer. We will once again judge how unusual the study's result of 14/16 would be under this assumption.

Group 1

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

Easton: 7

Brennan: 9

Brodey: 7

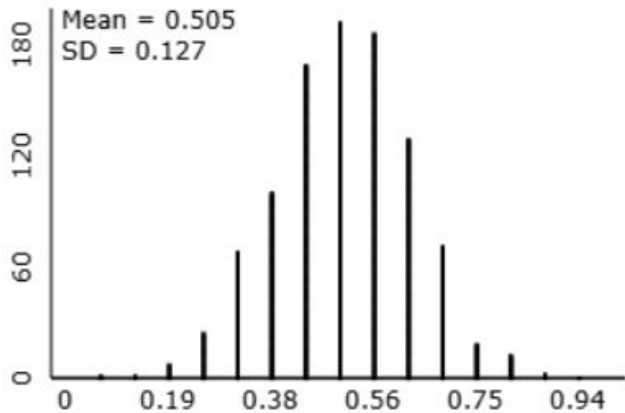
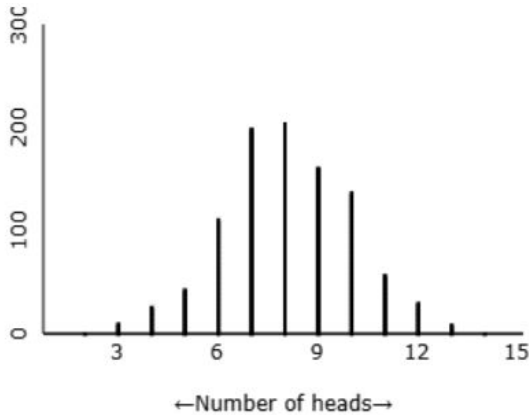
Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

Easton: 9,10,10,9

Brennan: 9,6,9,8 - Was not the same every time.

Brodey: 6, 10, 7, 10

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.



Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

With the assumption that the null hypothesis is true, the average proportion of heads is .505 with a standard deviation of 0.127. In the skeptic’s model it is very unlikely that 14/16 babies chose the helper without bias because it is in the 0.875 proportion of our sample graph and it looks like we only have 1 result there out of 1000 which represents a 0.1% chance.

Extra slide for you, just in
case

Group 2

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

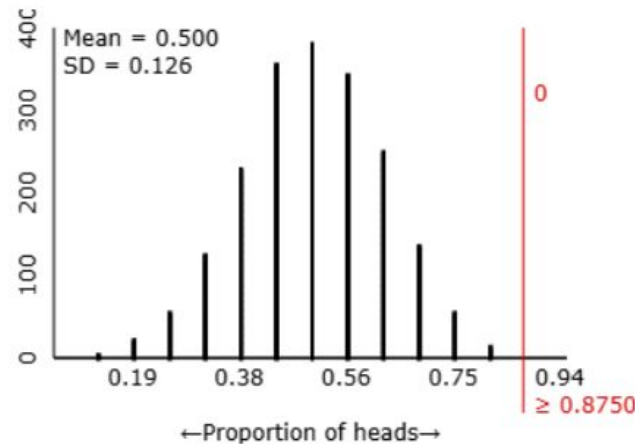
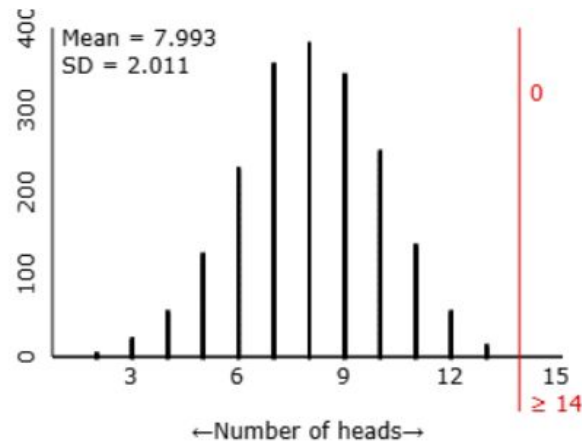
Colin: 11 heads, 6 heads, 12 heads, 3 heads

Sophia: 11 heads, 6 heads, 9 heads, 7 heads

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

No it was not the same every time.

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.



Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

The null model shows a normal (bell-shaped) distribution centered around 0.5 with a standard deviation of 0.126. This distribution suggests that we can reject the null hypothesis because there were **no** simulations that had a proportion ≥ 0.875 (which is the proportion of the experimental data).

Extra slide for you, just in
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Group 3

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.

Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

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Group 4

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

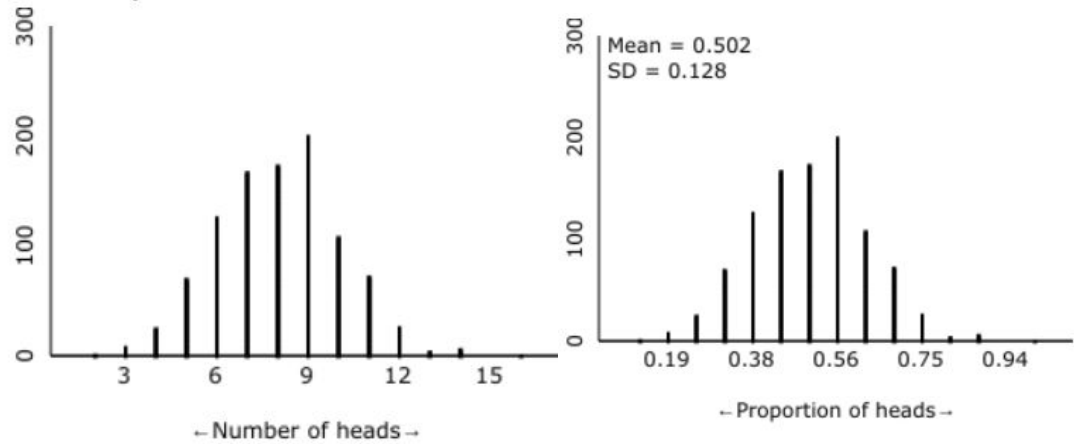
Itunu: 6, 7, 10, 12, 11

Emily: 9, 12, 9, 10, 10

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

No, it was not the same every time.

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.



Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

The dotplot shows 1000 simulated experiments, where the proportion of heads out of 16 coins was recorded for each simulation. The graph shows a normal distribution. The average proportion of heads was .502, give or take .128. When assuming our test statistic was purely by chance, this result would only occur 8/1000 times, which is unusual. Therefore, the probability of the result occurring due to chance is very small at .008, which means we have enough evidence to reject null hypothesis. In other words, this is evidence that babies did not pick the helper by chance.

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Group 5

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

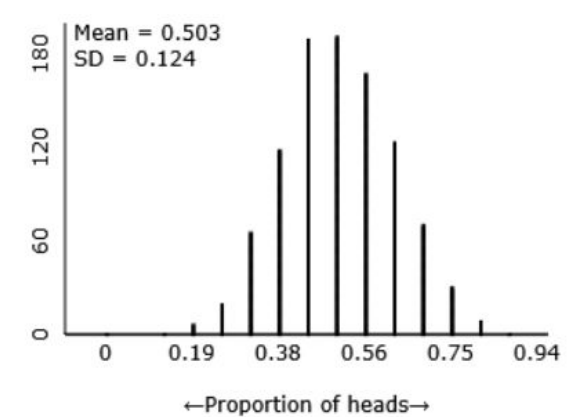
In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

8

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

6, 9, 10, 9

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.



Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

This graph represents the proportion of babies who chose helper in our simulated results when assuming the null hypothesis is true. The data appears to be normal, with a mean of 0.503 and a standard deviation of 0.124, with most of our data falling between .38 and .66 roughly. When comparing the data from this to our sample statistic, we can see that our statistic of 0.875 (14/16) seems very unusual, having a probability of 0.002 (2/1000). Thus we can reject the null hypothesis as it is very unlikely that random variation is the reason for babies picking helper at this proportion.

Extra slide for you, just in
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Group 6

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

Stone - 6 infants picked the helper. (6 heads)

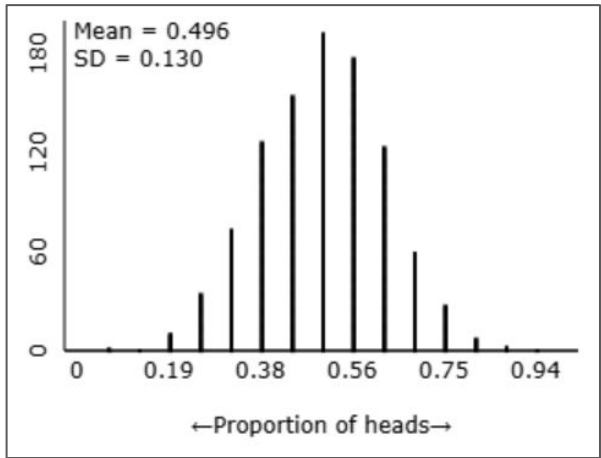
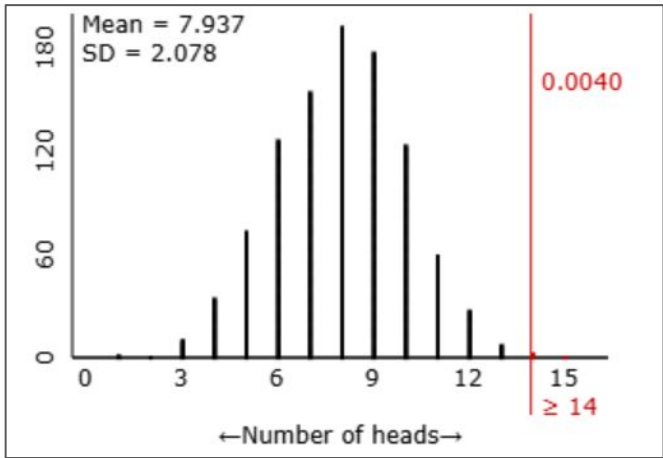
Jonathan - 9

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

Stone - 11, 10, 6, 8. No, I got different numbers almost every time.

Jonathan - 6, 5, 7, 9. No, it was not.

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.



Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

The graph shows a normal bell curve distribution. A mean of 7.937 (counts) shows that roughly half of the 1000 times we flipped our 16 coins we averaged a mean of 0.496 putting us very close to the null hypothesis assumption of 0.5. This distribution suggests that the alternate hypothesis should be affirmed and the babies were indeed able to pick out the helper 14/16 times. In our simulated runs we only had 4/1000 times in which we achieved the same or similar results so the odds of the babies doing the same at random chance is only 0.4%. Therefore, we can reject the null hypothesis, but more testing would be required to determine if the alternate hypothesis is true due to the small sample size.

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Group 7

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

Daniel: 11 heads

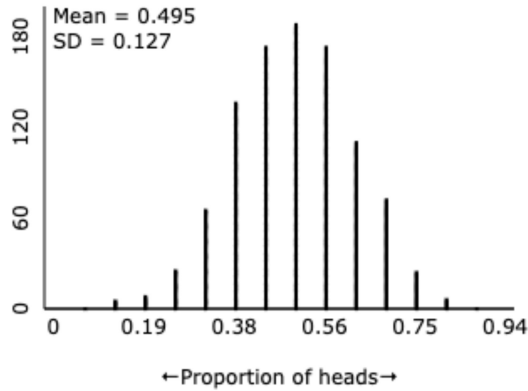
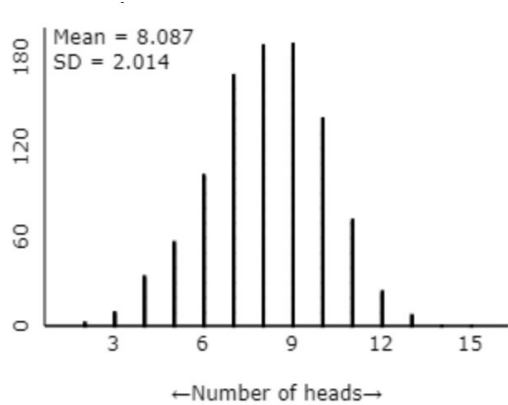
Moyed: 5 heads

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

Daniel: Moyed:

- | | |
|------|-------|
| 1)8 | 1) 8 |
| 2)10 | 2) 11 |
| 3)7 | 3) 9 |
| 4)5 | 4)2 |

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.



Please do proportions in your graph, not number

Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

The graph is a normal bell shaped distribution, the number of times heads was flipped was between 2 - 15, and an average heads flipped of 8. With a test statistics of 0.875 on the graph, there are nearly zero other values as extreme as it indicating a low p-value so we can reject the null hypothesis, making our model a plausible explanation for the experimental results

Extra slide for you, just in
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Group 8

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.

Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

Extra slide for you, just in
case

Group 9

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

Alexis: 7 heads

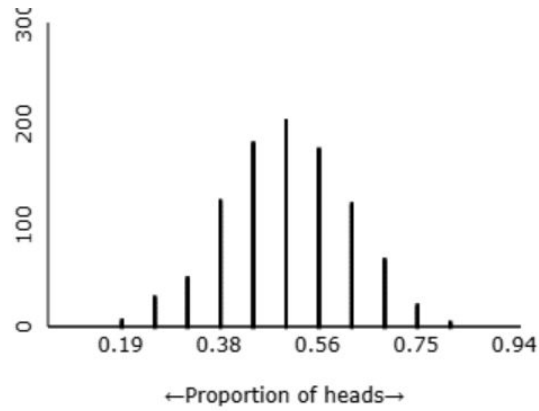
Landon: 12 heads

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

Alexis: 8, 10, 10, 12, not the same every time.

Landon: 9 heads, 10 heads, 11 heads, 8 heads. It was not the same everytime

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.



Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

The graph appears to take on a normal bell curve shape, with a center of 0.5 with a deviation of .119 between points. The number of points that assumes that a proportion greater than or equal to .875 when the null hypothesis is true is exactly zero, meaning that there is a very likely chance that the null hypothesis is rejected.

Extra slide for you, just in
case

Group 10

Use the following applet to simulate a single trial of 16 coin flips.

<http://www.rossmanchance.com/applets/OneProp/OneProp.htm>

- Keep “Probability of heads” set at 0.5.
- Change “Number of tosses” to 16.
- Keep “Number of repetitions” at 1 for now.
- Click “Draw samples”.

In this simulation, how many infants picked the helper (i.e. how many heads did you get)? Have each person in the group report to make certain that everyone got the applet working. Record results for each member of the group below.

Jonah: 10 heads, 6 tails

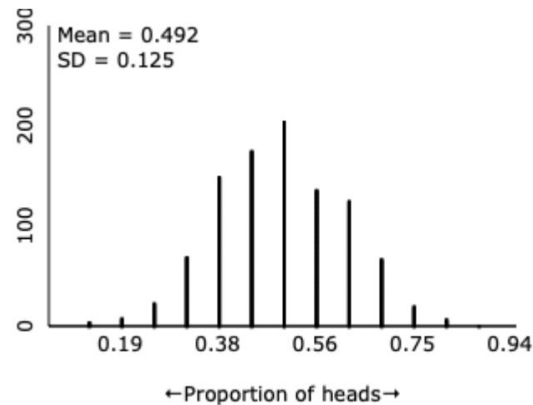
Jose: 8 heads 8, Tails

Uncheck the “Show animation” box and click “Draw samples” four more times. Record how many heads you got in these four trials. Was it the same every time?

Jonah: 6 heads, 10 tails/ 6 heads, 10 tails/ 11 heads, 5 tails/ 4 heads, 12 tails

Jose: 7 heads 9 tails, 6 heads 10 tails, 10 heads 6 tails, 13 heads 3 tails

Now change “Number of repetitions” to 995 and click “Draw samples” so that you have a total of 1000 repetitions recorded in the applet. Have one person take a screenshot of their histogram from the applet and paste it below.



Please do proportions in your graph, not number

Describe the distribution of the number of heads in your repetitions under the null model. As always, describe the center, spread, and shape of the distribution. What does this distribution suggest about whether our precise version of the skeptic’s model is a plausible explanation for the experimental results?

The graph represents 1000 coin flip simulations where each dot represents the proportion of heads flipped in 16 coins. It seems to be bell shaped, with the average amount of times it flips heads being 0.49. The number of times that heads was flipped is between 0.125 and 0.875. The data above shows how the explanation that we have come up with is true and it is plausible.

Extra slide for you, just in
case